BUDT758L: Pricing Optimization and Revenue Management - Copa Cruise Team Project

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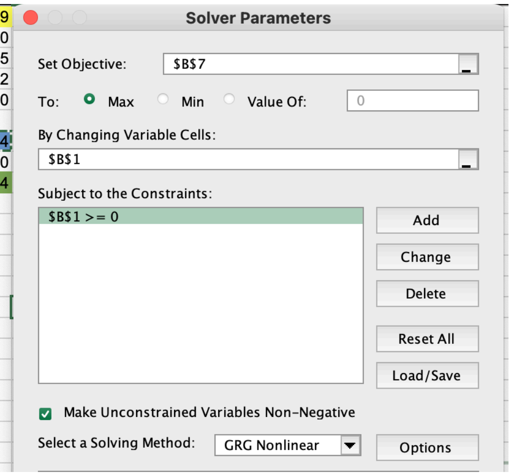
1.

a. The price per person that maximizes expected revenue for Copa is $101.99. The expected revenue per transaction at this price is calculated as:
101.99 * 0.703 * 50 = 3,588.42

So, the expected revenue per event is $3,588.42.

Price per person	101.99
W(p)	0.70
Revenue	5099.55
Expected revenue	3588.42
no of events	7155.00
Total exp. Revenue	25675112.94
Rvenue last year	16927650.00
Difference	8747462.94



b. If Copa maintains the same number of inquiries and sets the price at **$101.99** per person (as determined earlier), the total expected revenue for the company would be around **$25,675,112**.

When compared to the revenue earned in 2017, which was **$16,927,650**, this would result in an increase of about **$8,747,462**. In other words, by using the optimal price of **$101.99**, Copa could boost its annual revenue by nearly **$8.7 million**, assuming the number of inquiries stays the same.

2.

a. For pricing optimization, we segmented the data based on three distinct criteria: **event type**, **location**, and **booking date**. These criterias were chosen because they capture essential aspects of customer behavior and willingness to pay, which helps Copa Cruises develop a more tailored pricing strategy.

- **Event Type**: Events were segmented by type: **Wedding**, **Business**, and **Private**. Each of these events has unique customer perceptions of value and sensitivity to pricing. For instance, weddings generally carry a higher emotional and financial investment, making customers more willing to pay a premium.

Unique Type	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
Wedding	-4.1866374474	0.03244362938	=SUMIF($D$2:$D$7156,O2,$M$2:$M$7156)	102.871882247	=1/(1+EXP(P2+Q2*S2))	=S2*50	=U2*T2	=COUNTIF(D2:D7156,O2)
Business	-4.2436110503	0.03260299417	=SUMIF($D$2:$D$7156,O3,$M$2:$M$7156)	103.596062533	=1/(1+EXP(P3+Q3*S3))	=S3*50	=U3*T3	=COUNTIF(D3:D7157,O3)
Private	-4.3130026982	0.03441756846	=SUMIF($D$2:$D$7156,O4,$M$2:$M$7156)	99.5577433552	=1/(1+EXP(P4+Q4*S4))	=S4*50	=U4*T4	=COUNTIF(D4:D7158,O4)
							Total Revenue	=SUMPRODUCT(V2:V4,W2:W4)
							Revenue Diff	=W5-'1a,b'!B8

- **Location**: We also segmented the data by **location**. Each city or region where Copa operates has its own market dynamics, with different demand elasticities and cost structures. For example, events in **New York City** typically command higher prices due to the area's elevated costs and customer expectations.

Location-ID	Location	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
1	Boston	-3.377826564	0.02714392	=SUMIF($B$2:$B$7156,O2	102.87188224781	=1/(1+EXP(	=T2*50	=V2*U2	=COUNTIF(B2:B7156,O2)
2	NYC	-3.957813673	0.03096982	=SUMIF($B$2:$B$7156,O3	103.59606253379	=1/(1+EXP(	=T3*50	=V3*U3	=COUNTIF(B3:B7157,O3)
3	Washington DC	-4.537525029	0.03537824	=SUMIF($B$2:$B$7156,O4	99.557743355277	=1/(1+EXP(	=T4*50	=V4*U4	=COUNTIF(B4:B7158,O4)
4	Baltimore	-4.816930157	0.03654102	=SUMIF($B$2:$B$7156,O5	99.557743355277	=1/(1+EXP(	=T5*50	=V5*U5	=COUNTIF(B5:B7159,O5)
5	Norfolk VA	-4.274519024	0.03424432	=SUMIF($B$2:$B$7156,O6	99.557743355277	=1/(1+EXP(	=T6*50	=V6*U6	=COUNTIF(B6:B7160,O6)
6	Philadelphia	-3.922968470	0.03112700	=SUMIF($B$2:$B$7156,O7	99.557743355277	=1/(1+EXP(	=T7*50	=V7*U7	=COUNTIF(B7:B7161,O7)
7	New Jersey	-4.615999509	0.03572935	=SUMIF($B$2:$B$7156,O8	99.557743355277	=1/(1+EXP(	=T8*50	=V8*U8	=COUNTIF(B8:B7162,O8)
8	Seattle	-4.148599386	0.03290862	=SUMIF($B$2:$B$7156,O9	99.557743355277	=1/(1+EXP(	=T9*50	=V9*U9	=COUNTIF(B9:B7163,O9)
9	National Harbor MD	-4.794189136	0.03671512	=SUMIF($B$2:$B$7156,O1	99.557743355277	=1/(1+EXP(	=T10*50	=V10*U10	=COUNTIF(B10:B7164,O10)
10	Miami	-4.051241999	0.03100428	=SUMIF($B$2:$B$7156,O1	99.557743355277	=1/(1+EXP(	=T11*50	=V11*U11	=COUNTIF(B11:B7165,O11)
								Total	=SUMPRODUCT(W2:W11,X2:
								Revenue Diff	=X12-'1a,b'!B8

- **Booking Date**: Finally, segmentation was done by **booking window**, referring to how far in advance the event was booked. Customers booking closer to the event date may be less sensitive to price, while those booking further in advance might prioritize securing a good deal.

Booking date(Categorized)	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
151-180 days	-3.328290644	0.03080739	=SUMIF($D$2:$D$7156,S2	99.401573651485	=1/(1+EXP(	=W2*50	=Y2*X2	=COUNTIF($D$2:$D$7156,S2)
31-60 days	-6.596441621	0.04263728	=SUMIF($D$2:$D$7156,S3	104.70182609778	=1/(1+EXP(	=W3*50	=Y3*X3	=COUNTIF($D$2:$D$7156,S3)
More than 180 days	-3.703582280	0.03234506	=SUMIF($D$2:$D$7156,S4	105.91222162190	=1/(1+EXP(	=W4*50	=Y4*X4	=COUNTIF($D$2:$D$7156,S4)
91-120 days	-4.017543342	0.03331048	=SUMIF($D$2:$D$7156,S5	103.69781545014	=1/(1+EXP(	=W5*50	=Y5*X5	=COUNTIF($D$2:$D$7156,S5)
61-90 days	-5.029135562	0.03799743	=SUMIF($D$2:$D$7156,S6	103.19274526391	=1/(1+EXP(	=W6*50	=Y6*X6	=COUNTIF($D$2:$D$7156,S6)
121-150 days	-4.093816315	0.03601460	=SUMIF($D$2:$D$7156,S7	104.06744233834	=1/(1+EXP(	=W7*50	=Y7*X7	=COUNTIF($D$2:$D$7156,S7)
30 days or less	-7.978314839	0.04738214	=SUMIF($D$2:$D$7156,S8	105.76438790100	=1/(1+EXP(	=W8*50	=Y8*X8	=COUNTIF($D$2:$D$7156,S8)
							Total	=SUMPRODUCT(Z2:Z8,AA2:A
							Revenue Diff	=AA9-'1a,b'!B8

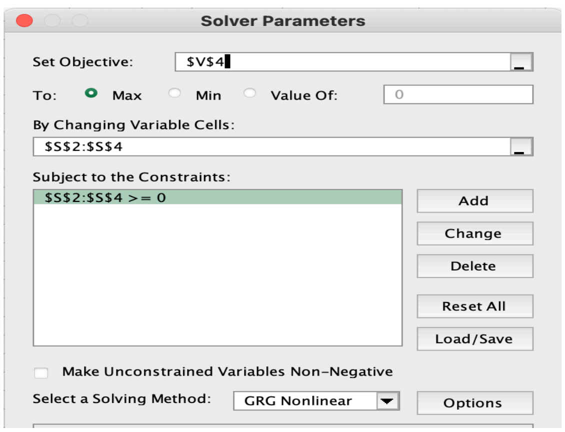
These segmentation criteria allow Copa Cruises to target specific segments effectively and optimize revenue by adjusting pricing to each segment's characteristics.

b. Based on the analysis, the revenue-maximizing prices and expected revenues for each segment are as follows:

By Event Type:

- **Wedding**: The optimal price is **$102.87**, with an expected revenue of **$3,602.46** per transaction.
- **Business**: The optimal price is **$103.60**, with an expected revenue of **$3,646.20** per transaction.
- **Private**: The optimal price is **$99.56**, with an expected revenue of **$3,525.14** per transaction.

Unique Type	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
Wedding	-4.19	0.03	-1360.65	102.87	0.70	5143.59	3602.46	2399.00
Business	-4.24	0.03	-1355.41	103.60	0.70	5179.80	3646.20	2375.00
Private	-4.31	0.03	-1292.63	99.56	0.71	4977.89	3525.14	2381.00
							Total Revenue	25695407.54
							Revenue Diff	8767757.54

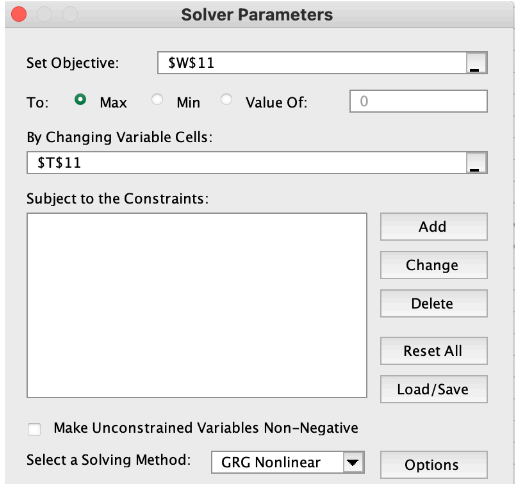


These prices reflect the unique characteristics of each event type, ensuring Copa Cruises captures the highest potential revenue per event type.

By Location:

- **Boston**: Optimal price is **$102.87**, with expected revenue of **$3,303.87** per transaction.
- **New York City**: Optimal price is **$103.60**, with expected revenue of **$3,517.40** per transaction.
- **Washington DC**: Optimal price is **$99.56**, with expected revenue of **$3,654.09** per transaction.
- **Other Locations**: Prices hover around **$99.56** with slight variation in expected revenue between **$3,500 and $3,800** per transaction depending on the location.

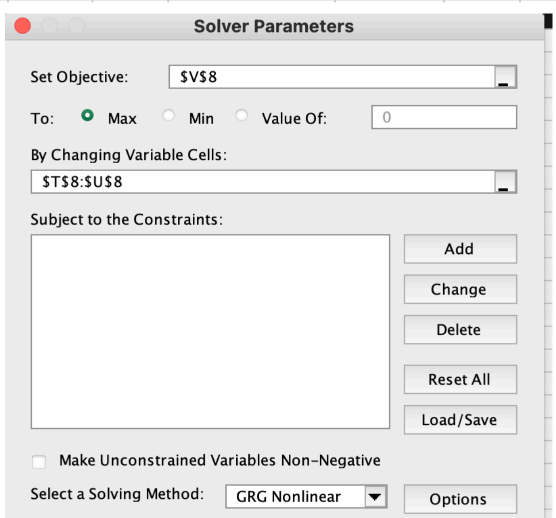
Location-ID	Location	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
1.00	Boston	-3.38	0.03	-409.70	102.87	0.64	5143.59	3303.87	708.00
2.00	NYC	-3.96	0.03	-418.55	103.60	0.68	5179.80	3517.40	726.00
3.00	Washington DC	-4.54	0.04	-374.77	99.56	0.73	4977.89	3654.09	685.00
4.00	Baltimore	-4.82	0.04	-412.05	99.56	0.76	4977.89	3806.92	748.00
5.00	Norfolk VA	-4.27	0.03	-390.59	99.56	0.70	4977.89	3503.20	722.00
6.00	Philadelphia	-3.92	0.03	-411.97	99.56	0.70	4977.89	3460.09	728.00
7.00	New Jersey	-4.62	0.04	-395.25	99.56	0.74	4977.89	3695.95	710.00
8.00	Seattle	-4.15	0.03	-384.55	99.56	0.71	4977.89	3510.52	691.00
9.00	National Harbor MD	-4.79	0.04	-398.19	99.56	0.76	4977.89	3770.65	730.00
10.00	Miami	-4.05	0.03	-410.90	99.56	0.72	4977.89	3604.12	702.00
								Total	25624223.34
								Revenue Diff	8696573.34



By Booking Date(in advance):

- **<30 days**: The optimal price is **$105.76**, with an expected revenue of **$5,029.41** per transaction.
- **31-60 days**: The optimal price is **$104.70**, with an expected revenue of **$4,680.16** per transaction.
- **61-90 days**: The optimal price is **$103.19**, with an expected revenue of **$3,878.86** per transaction.
- **91-120 days**: The optimal price is **$103.70**, with an expected revenue of **$3,303.92** per transaction.
- **121-150 days**: The optimal price is **$104.07**, with an expected revenue of **$3,047.17** per transaction.
- **151-180 days**: The optimal price is **$99.40**, with an expected revenue of **$2,813.60** per transaction.
- **> 180 days**: The optimal price is **$105.91**, with an expected revenue of **$3,013.30** per transaction

Booking date(Categorized)	a	b	Max(ln(LE))	Price	W(p)	Revenue	Exp Revenue	Events
151-180 days	-3.33	0.03	-614.49	99.40	0.57	4970.08	2813.60	1226.00
31-60 days	-6.60	0.04	-654.59	104.70	0.89	5235.09	4680.16	1191.00
More than 180 days	-3.70	0.03	-392.25	105.91	0.57	5295.61	3013.30	758.00
91-120 days	-4.02	0.03	-613.81	103.70	0.64	5184.89	3303.92	1154.00
61-90 days	-5.03	0.04	-657.14	103.19	0.75	5159.64	3878.86	1211.00
121-150 days	-4.09	0.04	-574.14	104.07	0.59	5203.37	3047.17	1192.00
30 days or less	-7.98	0.05	-214.93	105.76	0.95	5288.22	5029.41	423.00
							Total	25577314.47
							Revenue Diff	8649664.47



Each of these prices and expected revenues is aligned with the customer's booking behavior and location, enabling Copa Cruises to maximize its returns based on demand patterns and customer preferences.

c. The annual expected revenue for Copa for each segment:

By Event Type:

- **Wedding**: With 2,399 events annually, the expected annual revenue is **$8,641,090.26**.
- **Business**: With 2,375 events annually, the expected annual revenue is **$8,665,725.25**.
- **Private**: With 2,381 events annually, the expected annual revenue is **$8,390,994.30**.

In total, the expected revenue using optimal pricing for event type segmentation would be **$25,965,407.54**, representing an increase of **$9,037,757.54** compared to last year's total revenue of **$16,927,650**.

By Location:

The expected annual revenue using the optimal price for location segmentation results in **$25,622,423.34**, which is **$8,694,773.34** higher than last year's total revenue.

By Booking Date:

The expected annual revenue based on booking date segmentation is **$25,577,315.45**, showing a potential increase of **$8,649,665.45** compared to 2017's total revenue.

In conclusion, applying segmentation allows Copa Cruises to implement a more nuanced pricing strategy that reflects the specific behaviors and preferences of different customer segments. Whether by event type, location, or booking date, optimizing pricing for each segment can significantly enhance the company's revenue. Compared to 2017, these segmentation-based strategies could potentially generate an additional **$8.7 million** to **$9 million** annually.

3. Yes, the differences in cost would affect the segmentation criteria. Currently, we have taken into consideration the expected revenue for each segment. However, if we have the cost factor, we can calculate the contribution margin and subsequently the expected profits for each segment. This can enable us to select the most attractive segment and implement tailored pricing strategies. For example,

- Early booking vs late booking can be a factor for difference in costs. Last minute booking can lead to increased costs for the company.
- Different locations may have different labor cost, rental costs, etc
- Different event types can have different costs, for example a wedding might have significantly higher costs as compared to a private event.

4. Yes, the analysis will change if we have different group sizes, as per unit price, per unit revenue and total revenue will be different for each customer segment. This will affect our pricing strategy and help us identify the most attractive segment based on our goal i.e. revenue maximization or profit maximization.